
SGERA: Stein-Guided ECG-Report Alignment for ECG Representation Learning

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Abstract

Electrocardiogram (ECG) representation learning via ECG-report alignment is often hindered by the inherent structural and statistical divergence between signals and natural language. Existing methods struggle to bridge this gap with simple contrastive objectives but struggle with distribution dependencies between heterogeneous features. To address this, we propose **SGERA** (Stein-Guided ECG-Report Alignment), which leverages the unique properties of Stein kernels to provide a more rigorous geometric alignment in the latent space: **instance-level** alignment via a Stein-RBF kernel enforces pairwise consistency between ECG and report embeddings, and **distribution-level** alignment via a Stein-Score kernel captures higher-order interactions for global alignment. Furthermore, we introduce an ECG-Report matching task with a Hard Sample Mining strategy to refine discriminative boundaries. Experiments across three public datasets demonstrate that SGERA significantly outperforms state-of-the-art SSL methods in zero-shot classification, linear probing, and transfer learning, proving the superiority of Stein-guided alignment in handling complex medical modalities. Code is available at <https://github.com/cccccj-03/SGERA/tree/main>.

1. Introduction

Electrocardiograms (ECGs) are a fundamental tool for monitoring cardiac electrical activity and diagnosing a wide

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range of cardiovascular conditions (Huang et al., 2022a;b). However, conventional ECG interpretation heavily relies on the expertise of trained clinicians, making large-scale and consistent analysis labor-intensive and subjective. Recent advances in deep learning have significantly improved the automatic modeling of ECG signals, enabling data-driven representation learning that facilitates accurate and scalable intelligent diagnosis (Siontis et al., 2021; Yuan et al., 2025).

Due to the fact that large-scale data annotation requires significant time and human resource costs, self-supervised learning (SSL), which aims to learn informative ECG representations without labels, offers a feasible solution (Krishnan et al., 2022; Liu et al., 2025). Traditional SSL methods focus on single-modal learning and rely on either generative (Hu et al., 2023; Na et al., 2024; Jin et al., 2025) or contrastive methods (McKeen et al., 2024; Chen et al., 2024; Yuan et al., 2025). However, these methods focus solely on ECG signals while overlooking complementary clinical knowledge. To adopt clinical knowledge into ECG representation learning, ECG-report alignment methods are proposed (Liu et al., 2024; Yu et al., 2024; Hung et al., 2025). With the help of clinical reports, ECG representation learning shows great performance on downstream tasks.

Despite advances in ECG-report alignment methods, most rely on a CLIP-style (Radford et al., 2021) objective that enforces instance-level similarity between ECG and report embeddings. From a statistical view, this formulation implicitly assumes that embeddings from the two modalities are sampled from compatible distributions. However, ECG signals and clinical reports originate from fundamentally different sources, leading to pronounced discrepancies in their underlying feature distributions and semantic abstraction levels. Therefore, direct instance-wise alignment may introduce biased or unstable supervision, as high similarity does not necessarily imply distributional consistency across modalities. These limitations underscore the need for alignment mechanisms that go beyond point-wise matching and instead explicitly measure and reduce distributional discrepancies between heterogeneous ECG and report embeddings.

To address this gap, we propose **SGERA**, Stein-Guided ECG-Report Alignment, which leverages Stein kernel-

based alignment to minimize the discrepancy between heterogeneous ECG and report representations. Specifically, SGERA introduces two key innovations: (i) an adaptive Stein-RBF kernel-based instance alignment loss that replaces CLIP-style contrastive loss to perform stable instance-level alignment, dynamically adjusting kernel bandwidth to adopt cross-modal distribution shifts; and (ii) a Stein-Score kernel distributional alignment loss that explicitly captures higher-order distributional discrepancies between ECG and report embeddings, enabling distribution-level alignment without requiring explicit density estimation. By integrating instance kernel similarity with distribution-aware supervision, SGERA provides a theoretically grounded and robust framework for ECG-report representation learning. Moreover, we also introduce an ECG-Report Matching task to enhance instance discrimination. We conduct experiments on three public ECG datasets across multiple tasks to verify our proposed SGERA, demonstrating its superiority over existing ECG-Report alignment methods.

Our main contributions are summarized as follows:

- We first identify and analyze the theoretical limitations of existing CLIP-style ECG-report alignment methods from a distributional perspective, showing that direct instance-level contrastive alignment is insufficient for handling the inherent heterogeneity between ECG signals and clinical reports.
- We propose **SGERA**, a novel Stein-guided ECG-report alignment framework that formulates cross-modal representation learning as joint instance-level and distribution-level modeling. SGERA introduces an adaptive RBF Stein kernel for stable instance-level alignment and a score-based Stein kernel for principled distribution-level alignment. An ECG-Report Matching task with hard sample mining is also designed.
- We conduct extensive experiments on three public ECG datasets across multiple downstream tasks, including zero-shot and supervised settings, demonstrating that SGERA consistently outperforms existing ECG-report alignment methods and learns more robust and effective representations for downstream classification.

2. Related Work

2.1. Multi-modal ECG Representation Learning

Multi-modal ECG Representation Learning aims to adopt clinical reports to ECG representation learning for better clinical information extraction. MERL (Liu et al., 2024) first proposes a CLIP-style framework that aligns ECG signals with clinical text reports for zero-shot classification. C-MELT (Pham et al., 2024) jointly employs masked generative objectives for ECG and text, and integrates cross-modal

contrastive learning to align the two modalities. Furthermore, DERI (Chen et al., 2025) proposes a multiple alignment and reconstruction framework for deep ECG-report interaction, which enables better clinical information learning. MELP (Wang et al., 2025a) designs a multi-scale framework to conduct hierarchical alignment of ECG-report pairs. Despite these advances, existing methods largely ignore the distributional mismatch induced by the inherent heterogeneity between ECG signals and clinical reports, leading to suboptimal alignment. Moreover, noise and sparse correspondence in clinical reports further impede the learning of clinically salient ECG representations.

2.2. Stein-Guided Methods

Stein Variational Gradient Descent (SVGD) (Liu & Wang, 2016) provides a principled kernel-based approach for measuring and reducing distributional discrepancies through deterministic particle transformations. Stein-guided methods have proven their effectiveness in various alignment tasks (Liu et al., 2021; Xiao et al., 2025; Wang et al., 2025b). Compared to CLIP-style alignment, Stein-guided alignment operates at the distribution level, explicitly measuring discrepancies between embedding distributions through kernelized Stein operators (Wen et al., 2020). By incorporating score information and kernel-based interactions, Stein-guided alignment captures higher-order statistical differences beyond pointwise similarity and remains effective without explicit density estimation (Liu et al., 2021). To our knowledge, SGERA is the first to introduce Stein’s methods to multi-modal ECG representation learning, leveraging Stein kernels for aligning heterogeneous ECG signals and clinical reports, where distribution gap are prevalent.

3. Method

We first present the preliminaries by analyzing the limitations of CLIP-style alignment for ECG-report pairs. We then introduce Stein kernel-based alignment and provide theoretical analysis to justify its advantages in handling heterogeneous representations. Finally, we describe SGERA, a Stein-guided alignment framework designed for robust ECG-report representation learning, as shown in Figure 1.

3.1. Preliminaries

We first formulate the general objective of CLIP-style ECG-report alignment methods. Given $x \in \mathcal{D}_E$ and $y \in \mathcal{D}_R$ regarding ECG signals and paired clinical reports, two separated encoders $f(\cdot)$ and $g(\cdot)$ will usually be used to project them to the shared latent space: $z_E = f(x)$, $z_R = g(y)$. Then, CLIP-style methods typically achieve instance-level alignment by comparing learning objectives with In-

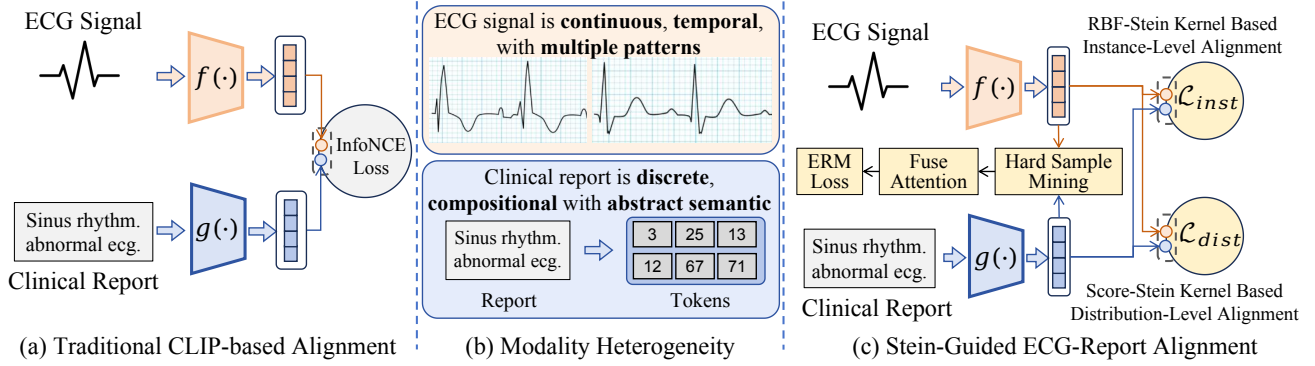


Figure 1. Overview of proposed SGERA. (a) Traditional CLIP-based ECG-report alignment relies on the InfoNCE loss, which enforces instance-level similarity while overlooking cross-modal distributional discrepancies. (b) Such discrepancies arise from modality heterogeneity: ECG representations capture dense, continuous patterns with fine-grained temporal and morphological details, whereas clinical reports provide sparse, high-level semantic abstractions. (c) To bridge this gap, we propose SGERA, which introduces Stein-guided dual-level alignment to explicitly account for distributional mismatch across modalities. In addition, a hard-negative ECG-Report Matching (ERM) task is incorporated as an auxiliary objective to enhance instance discrimination.

foNCE loss as Eq. 1:

$$\mathcal{L}_{CLIP} = -\mathbb{E}_{(x,y)} \left[\log \frac{\exp(\text{sim}(z_E, z_R)/\tau)}{\sum_{y'} \exp(\text{sim}(z_E, g(y'))/\tau)} \right], \quad (1)$$

where $\text{sim}(\cdot, \cdot)$ regards cosine similarity, y' regards unpaired reports within batch, and τ is the temperature parameter.

The optimization of this objective implies a key assumption:

Assumption 3.1. The ECG and report embeddings, $z_E \sim p_E(z)$ and $z_R \sim p_R(z)$, are assumed to be distributionally compatible in the shared latent space, ensuring that instance-level similarity faithfully reflects underlying cross-modal semantic correspondence.

However, Assumption 3.1 is often violated in the ECG-report alignment setting. ECG representations are derived from continuous, high-resolution physiological signals that encode fine-grained temporal and morphological patterns, whereas clinical reports provide discrete, high-level semantic summaries that may be incomplete, ambiguous, or noisy. These modality-specific generative processes induce substantially different embedding distributions, leading to mismatched support, scale, and higher-order statistics between $p_E(z)$ and $p_R(z)$. Consequently, high instance-level similarity does not necessarily imply distributional consistency, and enforcing pointwise alignment alone may introduce biased or unstable supervision.

In addition, according to Eq. 1, we can obtain the gradient of ECG representation as Eq. 2:

$$\begin{aligned} \nabla_{z_E} \mathcal{L} &= \frac{1}{\tau} \left(\sum_j p_j z_R^j - z_R^+ \right), \\ p_j &= \text{softmax}(\langle z_E, z_R^j \rangle / \tau), \end{aligned} \quad (2)$$

Therefore, the resulting gradients are dominated by a limited

set of report embeddings z_R within each mini-batch, yielding updates that are inherently local and instance-driven rather than distribution-aware. In the ECG-report setting, where the two modalities arise from fundamentally different sources, such gradients can be highly sensitive to semantic noise and incompleteness in clinical reports. As a consequence, instance-level alignment may amplify distribution mismatch instead of mitigating it, producing unstable or biased supervision. This risks overfitting superficial textual co-occurrences while distorting the geometric structure of ECG representations, ultimately undermining the learning of physiologically meaningful features.

This observation motivates the need for a hierarchical alignment mechanism that reconciles both local and global modal discrepancies. Specifically, while instance-level alignment remains essential for capturing fine-grained, pairwise semantic correspondences, it must be augmented by distribution-level alignment to bridge the broader structural gaps between heterogeneous modalities. We address this dual requirement through a Stein-guided framework, leveraging Stein-RBF kernels for robust point-to-point mapping and Stein-Score kernels for global manifold synchronization.

3.2. Stein Kernel-based Alignment

To explicitly address distribution gaps in ECG-report alignment, we propose a Stein Kernel-based alignment framework that unifies instance-level and distribution-level. Our method consists of two components: (i) an adaptive RBF-Stein kernel that replaces CLIP-style contrastive loss to perform stable instance-level alignment under cross-modal distribution shifts; and (ii) a score-based Stein kernel alignment that captures higher-order distributional discrepancies between ECG and report embeddings without explicit density estimation. Together, these components provide a principled

and robust alternative to pointwise contrastive alignment for heterogeneous ECG-report representation learning. More details can be found in Appendix A.

RBF-Stein Kernel for Instance-Level Alignment. We first focus on stabilizing instance-level alignment between ECG-report pairs under cross-modal distribution shifts. Unlike CLIP-style methods that rely on cosine similarity, we adopt a Stein-guided kernel formulation derived from an RBF kernel, but without the exponential mapping. Concretely, given ECG embeddings z_E^i and report embeddings z_R^j , we define a distance-based kernel interaction as Eq. 3:

$$K_{ij}^{rbf} = -\frac{\|z_E^i - z_R^j\|^2}{2\sigma^2}, \quad (3)$$

where σ denotes the kernel bandwidth. This form preserves the relative geometry of embedding pairs while avoiding the saturation and vanishing gradients induced by the exponential function in conventional RBF kernels.

Given the kernel matrix $K_{rbf} \in \mathbb{R}^{B \times B}$ computed from the RBF-Stein kernel, we define the instance-level alignment loss by replacing the cosine similarity. Concretely, the instance-level alignment loss is then defined as a symmetric cross-entropy over the kernel logits as Eq. 4:

$$\begin{aligned} \mathcal{L}_{rbf} = & \left(-\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(K_{ii}^{rbf}/\tau)}{\sum_{j=1}^N \exp(K_{ij}^{rbf}/\tau)} \right) + \\ & \left(-\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(K_{ii}^{rbf}/\tau)}{\sum_{j=1}^N \exp(K_{ji}^{rbf}/\tau)} \right) \end{aligned} \quad (4)$$

where τ is the temperature parameter. In addition, a KL divergence between the encoded Gaussian distributions of ECG and report embeddings is used to regularize the latent space, as Eq. 5:

$$\begin{aligned} \mathcal{L}_{KL} = & -\frac{1}{2} \left(\sum (1 + \log \sigma_E^2 - \mu_E^2 - \sigma_E^2) \right. \\ & \left. + \sum (1 + \log \sigma_R^2 - \mu_R^2 - \sigma_R^2) \right) \end{aligned} \quad (5)$$

where $\sigma_E, \mu_E, \sigma_R$, and μ_R represent the standard deviations and means of the Gaussian latent variables encoded from ECG and report embeddings, respectively, capturing the uncertainty and scale of the corresponding representations.

Compared to the gradient of $\mathcal{L}_{CLIP}$ for an ECG embedding z_E^i as Eq. 2, the Stein kernel formulation yields as Eq. 6:

$$\nabla_{z_E^i} \mathcal{L} \propto \sum_{j=1}^N (z_R^j - z_E^i) (\tilde{p}_{ij} - \delta_{ij}) \quad (6)$$

where $\tilde{p}_{ij}$ denotes the softmax probability over the Stein kernel similarities, and δ_{ij} is the Kronecker delta indicating the ground-truth correspondence between ECG z_E^i and

report z_R^j ($\delta_{ij} = 1$ if they form a positive pair, 0 otherwise). Unlike the CLIP gradient, which can be sensitive to extreme similarity values due to the exponential in the softmax, the Stein kernel gradient is linear in the squared distance, producing smoother and more stable updates under cross-modal distribution shifts. Therefore, the total loss used for instance-level alignment is illustrated in Eq. 7:

$$\mathcal{L}_{inst} = \mathcal{L}_{rbf} + \mathcal{L}_{KL}. \quad (7)$$

The complete gradient derivation process and loss function comparison can be found in Appendix A.

Score-Stein Kernel for Distribution-Level Alignment. To move beyond instance-level alignment, we introduce a score-based Stein kernel for distribution-level alignment. This formulation leverages Stein’s identity to compare distributions via their score functions, capturing higher-order discrepancies without explicit density estimation. As a result, it provides principled, distribution-aware supervision for aligning heterogeneous ECG and clinical report representations.

Specifically, z_E and z_R are parameterized as a Gaussian latent variable, from which samples are drawn using the reparameterization trick (Cremer et al., 2018) as Eq. 8:

$$z_E' = \mu_E + \epsilon \odot \sigma_E, \quad z_R' = \mu_R + \epsilon \odot \sigma_R, \quad (8)$$

where $\epsilon \sim \mathcal{N}(0, I)$. The sampled embeddings are ℓ_2 -normalized and fed into modality-specific score networks to estimate the corresponding score functions $\nabla_z \log p(z)$ for ECG and report representations. Given sampled embeddings z , and their estimated score functions $s(z)$, we construct the score-based Stein kernel matrix $K_{\text{score}}(\cdot, \cdot)$ following Stein’s identity. This kernel captures higher-order distributional information by jointly modeling pairwise interactions between samples and their score functions, without requiring explicit density estimation. Based on the whole RBF kernel $k(x_i, y_j) = \exp(-\frac{\|x_i - y_j\|^2}{2\sigma^2})$ and the Stein gradient estimator (Liu & Wang, 2016; Wen et al., 2024), the resulting Stein kernel between x_i and y_j is given by Eq. 9:

$$\begin{aligned} K_{\text{score}}^{ij} = & k(x_i, y_j) s_x(x_i)^\top s_y(y_j) + \nabla_{x_i} k(x_i, y_j)^\top s_y(y_j) \\ & + \nabla_{y_j} k(x_i, y_j)^\top s_x(x_i) + \text{tr}(\nabla_{x_i y_j} k(x_i, y_j)), \end{aligned} \quad (9)$$

which jointly models interactions between samples and their score functions, serving as a key approximation for maximizing the embeddings’ differential entropy and thereby encouraging global uniformity in the latent space. To align distributions across modalities, we compute three Stein kernel matrices: (i) ECG-ECG, (ii) report-report, and (iii) ECG-report. We then minimize the squared cosine distance between these kernel matrices, as Eq. 10:

$$\mathcal{L}_{stein}(K_1, K_2) = 1 - \left(\frac{\langle K_1, K_2 \rangle}{\|K_1\|_F \|K_2\|_F} \right)^2, \quad (10)$$

where diagonal elements are removed to avoid trivial self-similarity. The final distribution-level alignment objective encourages both intra-modality consistency and cross-modality agreement by jointly minimizing $\mathcal{L}_{dist}$ as Eq. 11:

$$\mathcal{L}_{dist} = \mathcal{L}_{Stein}(K_E, K_R) + \mathcal{L}_{Stein}(K_E, K_{ER}) + \mathcal{L}_{Stein}(K_R, K_{ER}), \quad (11)$$

where K_E , K_R , and K_{ER} denote ECG, report, and cross-modal Stein kernel matrices, respectively. By enforcing alignment at the level of Stein-induced distributional geometry, this score-based objective complements instance-level alignment and provides robust supervision under modality heterogeneity, noise, and abstraction gaps.

Adaptive Kernel Bandwidth Update. To adapt the Stein kernel to the evolving embedding distribution, we update the bandwidth using a median-distance heuristic. Given sampled ECG and report embeddings z_E^i and z_R^j , the bandwidth is estimated as Eq. 12:

$$\sigma = \sqrt{\frac{\text{median}_{i,j} \|z_E^i - z_R^j\|_2^2}{2}} \cdot \gamma, \quad (12)$$

$$\sigma \leftarrow \text{clamp}(\sigma, \sigma_{min}, \sigma_{max})$$

where γ is a bandwidth scaling factor, σ_{min} and σ_{max} are used to constrain within a range to for numerical stability.

3.3. SGERA Framework

Based on the Stein kernel-based alignment above, we propose SGERA, a novel Stein-Guided ECG-Report Alignment framework. SGERA comprises three core components: modality-specific embedding learning, Stein-guided dual cross-modal alignment that operates at both the instance and distribution levels, and hard sample mining to learn fine-grained cross-modal features.

Modality-specific embedding learning. Given ECG-report pairs e_i, r_i , we adopt the 1D ResNet18 and the Med-CPT (Jin et al., 2023) as $f(\cdot)$ and $g(\cdot)$, the encoders for ECG signals and clinical reports respectively. After that, two non-linear projectors for ECG and text embedding are used to convert them into the same dimensionality as z_E and z_R . In addition, we adopt the same strategy as MERL (Liu et al., 2024) to obtain two views of ECG embedding $z_{E,1}$ and $z_{E,2}$ for feature enhancement.

Dual-level Alignment. Given learned ECG and text embedding z_E and z_R , we conduct Stein-guided cross-modal alignment from instance and distribution levels. Specifically, for instance-level, we adopt the instance-level alignment loss $\mathcal{L}_{inst}$ as Eq. 7 with z_E and z_R directly. For distribution-level, we adopt to MLPs as the estimated score functions $s_E(\cdot)$ and $s_R(\cdot)$ used for score-based Stein-Kernel K_{score}^{ij} as Eq. 9. We then use K_{score}^{ij} to obtain the distribution-level

alignment loss $\mathcal{L}_{dist}$ as Eq. 11. In addition, since the ECG embedding $z_{E,1}$ and $z_{E,2}$ are from the same distribution, we only conduct instance-level alignment on them, which the loss is regarded as $\mathcal{L}_{inst}^{uma}$.

Hard Sample Mining. While the main ECG-report alignment objective encourages global semantic consistency across modalities, it alone is insufficient to resolve fine-grained ambiguities caused by highly similar ECG morphologies or overlapping clinical descriptions. To this end, we introduce an auxiliary ECG-Report Matching (ERM) task that explicitly focuses on hard negative pairs. By mining intra-modal and cross-modal hard negatives, the ERM task enforces discriminative boundaries in the joint embedding space, preventing spurious alignments without interfering with the primary alignment objective.

Specifically, given z_E and z_R , we adopt a similarity-driven hard negative mining strategy, where negatives are sampled with probabilities proportional to their similarity scores. This allows the model to focus on semantically confusing examples that lie close to the decision boundary. Our strategy includes two views: intra-model and cross-model. For intra-modal hard negatives, we compute pairwise similarities within the batch based on z_E as Eq. 13:

$$s_{ij}^e = \frac{z_{i,E}^\top \cdot z_{j,E}}{\tau}, i \neq j, \quad (13)$$

and sample hard ECG negatives according to Eq. 14:

$$p_{ij}^e = \frac{\exp(s_{ij}^e)}{\sum_{k \neq i} \exp(s_{ik}^e)}. \quad (14)$$

A hard negative ECG embedding $\tilde{e}_i$ is then sampled from $e_j, j \neq i$ using p_{ij}^e . The same procedure is applied symmetrically to text embeddings to obtain hard text negatives $\tilde{t}_i$ with temperature τ_t . To further capture cross-modal confusions, we mine hard negatives across modalities by computing the cross-modal similarity.

In this way, given an ECG-report pair, we obtain four negative pairs. We then form a joint representation using a cross-modal fusion module on each pair (e, t) , as Eq. 15:

$$z = \text{FuseAttn}([e, t]), \quad (15)$$

where $\text{FuseAttn}(\cdot)$ denotes a self-attention-based fusion layer that captures cross-modal interactions. The fused representation z is then fed into a binary matching head $g(\cdot)$ to produce a matching score and obtain the ERM loss $\mathcal{L}_{erm}$ with cross-entropy loss.

Overall Pre-training. To pre-train our proposed SGERA, we jointly optimize three loss functions as Eq. 16:

$$\mathcal{L}_{total} = \mathcal{L}_{inst} + \mathcal{L}_{dis} + \mathcal{L}_{inst}^{uma} + \mathcal{L}_{erm}. \quad (16)$$

Table 1. Details on the number of samples in each split for dataset used.

# Samples	MIMIC-ECG	PTBXL-Super	PTBXL-Sub	PTBXL-Form	PTBXL-Rhythm	CPSC2018	CSN
Training	745,447	16,832	17,084	7,197	17,084	4,950	16,546
Validation	15,171	2,100	2,146	901	2,146	551	1,860
Testing	-	2,098	2,158	880	2,158	1,376	4,620

Transferring into Downstream Tasks. We transfer our pre-trained SGERA into downstream tasks as follow. For zero-shot classification, we directly used the learned z_E and adopt CKEPE (Liu et al., 2024) as the category prompts to obtain z_C as the same way as z_R . We then calculate the similarity between z_E and z_C for retrieval-based classification.

For fine-tuning to the linear probing setting, a trainable linear classification layer is added with z_E while all other parameters are frozen.

4. Experiments

4.1. Experimental Setup

4.1.1. PRETRAINING TASK.

We adopt the MIMIC-IV-ECG v1.0 dataset (Gow et al., 2023), including 800,035 ECG-report pairs recorded from 161,352 patients. All the ECG recordings are sampled at 500Hz for a duration of 10 seconds with their associated clinical reports. Following the same setting of MERL (Liu et al., 2024), we conduct preprocessing and obtain 760,618 ECG-text pairs for pretraining. The AdamW optimizer with an initial learning rate of $1e-3$ and a weight decay of $1e-8$ is adopted for 20 epochs pretraining with a cosine annealing learning rate scheduler. All experiments are conducted on 4 NVIDIA GTX 4090 GPUs with a batch size of 256. And γ is set as 1.0 while σ_{min} is 0.1 and σ_{max} is 10.

4.1.2. DOWNSTREAM TASKS.

After pretraining, the SGERA model is then evaluated across three public ECG benchmark datasets: PTB-XL (Wagner et al., 2020), CSN (Zheng et al., 2022), and CPSC2018 (Liu et al., 2018). We summarize the key dataset statistics in Table 1, and more details are provided in Appendix C. We adopt the same experimental configuration of MERL (Liu et al., 2024) for evaluation consistency.

PTB-XL contains 21,837 12-lead ECG samples, sampled at 500 Hz with a 10-second duration, from 18,885 subjects. According to the cardiac diagnostic annotation, the dataset can be stratified into four subgroups (super, sub, form, and rhythm) for granular evaluation.

CPSC2018 includes 6,877 12-lead ECG recordings sampled at 500 Hz, and each ECG recording is annotated with 9 categorical labels.

CSN contains 23,026 samples with 38 distinct diagnostic categories. All recordings are sampled at 500 Hz over 10-second intervals.

Tasks details. We adopt three tasks for evaluation, including zero-shot classification, linear probing, and distribution shift evaluation, following the setting of MERL (Liu et al., 2024). AUC (Area Under the Curve) is used as the evaluated metric of each model for ECG classification tasks. More implementation details about our experiments are reported in Appendix D for reproducibility.

4.2. Quantitative Results

4.2.1. ZERO-SHOT CLASSIFICATION

Following previous methods, we evaluate the zero-shot classification performance of SGERA with MERL (Liu et al., 2024), DERI (Chen et al., 2025), and MELP (Wang et al., 2025a). As shown in Table 2, SGERA achieves the best average performance (79.5%) among all methods, demonstrating superior zero-shot generalization across diverse ECG benchmarks. In particular, SGERA attains the best results on PTBXL-Form and CPSC2018, indicating its strong ability to capture both morphological characteristics and clinically diverse patterns. Unlike prior methods that exhibit dataset-specific advantages, SGERA performs best average AUC across datasets, highlighting its robustness to label granularity and dataset distribution shifts in zero-shot settings. SGERA slightly trails the leading baseline on the PTBXL-Rhythm and Sub tasks—a result of its design focus on generalizability over dataset-specific optimization. This trade-off mitigates representation bias, allowing SGERA to maintain more stable performance across heterogeneous data and achieve the best overall zero-shot generalization.

It should be noted that paired t-tests were performed to compare SGERA against MELP and DERI across all five folds/seeds. For the 0.5% average improvement over MELP shown in Table 2, the calculated p-value (specifically $p = 0.024$) indicates that the gain, while numerically small, is statistically consistent. In addition, regarding the 1.56% improvement over DERI, while the variance across tasks exists, SGERA consistently maintains the highest "lower bound" performance across subsets compared to baselines. This suggests that our Stein-based alignment provides a more stable representation even in tasks with high data sparsity. In medical ECG interpretation, even a 0.5%–1.5% improve-

Table 2. Zero-shot classification performance (AUC). The **best** and second best results are bold and underlined.

Methods	PTBXL-Super	PTBXL-Sub	PTBXL-Form	PTBXL-Rhythm	CPSC2018	CSN	Average
MERL (Liu et al., 2024)	74.2	75.7	65.9	78.5	82.8	74.4	75.3
D-BETA (Hung et al., 2025)	76.2	75.9	66.1	88.6	80.1	76.3	77.1
DERI (Chen et al., 2025)	79.8	76.7	67.6	85.6	85.9	76.8	78.7
MELP (Wang et al., 2025a)	76.2	81.2	69.1	85.4	84.2	77.6	79.0
SGERA (Ours)	<u>78.8</u>	<u>79.5</u>	71.3	83.3	87.1	<u>77.3</u>	79.5

 Table 3. Linear probing results (AUC). We bold the **best** results, and underline represents the second highest.

Method	PTBXL-Super			PTBXL-Sub			PTBXL-Form			PTBXL-Rhythm			CPSC2018			CSN		
	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%
SimCLR (Chen et al., 2020)	63.41	69.77	73.53	60.84	68.27	73.39	54.98	56.97	62.52	51.41	69.44	77.73	59.78	68.52	76.54	59.02	67.26	73.20
BYOL (Grill et al., 2020)	71.70	73.83	76.45	57.16	67.44	71.64	48.73	61.63	70.82	41.99	74.40	77.17	60.88	74.42	78.75	54.20	71.92	74.69
BarlowTwins (Zbontar et al., 2021)	72.87	75.96	78.41	62.57	70.84	74.34	52.12	60.39	66.14	50.12	73.54	77.62	55.12	72.75	78.39	60.72	71.64	77.43
MoCo-v3 (Chen & He, 2021)	73.19	76.65	78.26	55.88	69.21	76.69	50.32	63.71	71.31	51.38	71.66	74.33	62.13	76.74	75.29	54.61	74.26	77.68
SimSiam (Chen & He, 2021)	73.15	72.70	75.63	62.52	69.31	76.38	55.16	62.91	71.31	49.30	69.47	75.92	58.35	72.89	75.31	58.25	68.61	77.41
TS-TCC (Eldele et al., 2021)	70.73	75.88	78.91	53.54	66.98	77.87	48.04	61.79	71.18	43.34	69.48	78.23	57.07	73.62	78.72	55.26	68.48	76.79
CLOCS (Kiyasseh et al., 2021)	68.94	73.36	76.31	57.94	72.55	76.24	51.97	57.96	72.65	47.19	71.88	76.31	59.59	77.78	77.49	54.38	71.93	76.13
ASTCL (Wang et al., 2023)	72.51	77.31	81.02	61.86	68.77	76.51	44.14	60.93	66.99	52.38	71.98	76.05	57.90	77.01	79.51	56.40	70.87	75.79
CRT (Zhang et al., 2023)	69.68	78.24	77.24	61.98	70.82	78.67	46.41	59.49	68.73	47.44	73.52	74.41	58.01	76.43	82.03	56.21	73.70	78.80
STMEM (Na et al., 2024)	61.12	66.87	71.36	54.12	57.86	63.59	55.71	59.99	66.07	51.12	65.44	74.85	56.69	63.32	70.39	59.77	66.87	71.36
MERL (Liu et al., 2024)	82.39	86.27	88.67	64.90	80.56	84.72	58.26	72.43	79.65	53.33	82.88	88.34	70.33	85.32	90.57	66.60	82.74	87.95
HeartLang (Jin et al., 2025)	78.94	85.59	87.52	64.68	79.34	88.91	58.70	63.99	80.23	62.08	76.22	90.34	60.44	66.26	77.87	57.94	68.93	82.49
DERI (Chen et al., 2025)	85.46	89.84	90.52	73.50	80.60	85.52	62.53	72.51	84.37	65.44	83.66	92.34	79.45	89.40	93.45	77.93	87.86	91.93
MELP (Wang et al., 2025a)	<u>85.82</u>	87.61	87.87	79.22	84.40	87.46	63.41	76.71	83.30	88.83	94.65	96.91	88.54	91.75	<u>94.32</u>	78.25	84.83	90.17
SGERA	86.44	<u>89.81</u>	91.76	80.50	<u>83.94</u>	89.25	66.21	<u>75.30</u>	85.55	<u>80.68</u>	<u>91.05</u>	<u>93.68</u>	<u>80.91</u>	<u>91.13</u>	95.58	79.37	<u>86.44</u>	94.12

ment in AUC can be clinically meaningful when scaled to large-scale screening, where it translates to thousands of correctly identified rhythmic abnormalities.

4.2.2. LINEAR PROBING

We conducted linear probing using 1%, 10%, and 100% of the training data, and the results are shown in Table 3. Overall, SGERA demonstrates strong performance across datasets and label ratios, achieving the highest or second-highest average AUC in most settings. In particular, its advantage is more pronounced under low-label conditions (1%), suggesting that hard sample guidance helps improve feature representation when annotated data is scarce. However, SGERA does not outperform all baselines on every dataset or label ratio—for instance, MELP and DERI achieve slightly higher scores in certain high-label settings. These results indicate that while SGERA generally provides robust and competitive performance, its benefits are more prominent in low-resource scenarios while can be further improved in fine-grained ECG feature extractions.

4.2.3. DISTRIBUTION SHIFT

To evaluate the robustness to domain shift, we assess transfer learning performance on test datasets that, while sharing the same label space, exhibit a distribution shift relative to the pretraining data. Experimental results are shown in Table 4. SGERA achieves the best average AUC (83.34%) across all transfer settings, consistently outperforming both conventional self-supervised baselines and recent zero-shot

ECG foundation models. In particular, SGERA attains the strongest performance in three out of six transfer directions, including all transfers targeting PTBXL-Super and CPSC2018 to CSN scenarios, demonstrating robust generalization under significant distribution shift. This advantage stems from its structure-guided representation learning, which effectively mitigates dataset-specific representation bias while preserving transferable clinical semantics. Overall, these results validate SGERA as a robust and generalizable ECG representation under domain shift.

4.3. Analysis of Loss Landscape

To better understand the optimization behavior of our method, we analyze the loss landscape of the proposed Stein-Guided Alignment loss and remove the Stein kernel. This provides an intuitive perspective on the smoothness and curvature of the optimization surface, which are closely related to training stability and convergence speed. We visualize the loss landscape by perturbing the optimized parameters θ^* along two fixed random directions. Specifically, the loss is evaluated at $\theta^* + \alpha \mathbf{d}_1 + \beta \mathbf{d}_2$, where $\mathbf{d}_1$ and $\mathbf{d}_2$ are normalized directions and α, β control the perturbation magnitude. All evaluations are conducted on the same mini-batch to remove data-induced variance.

As shown in Figure 2, our proposed Stein kernel-based alignment loss (Figure (b)) exhibits more uniform and monotonic curvature compared to the original CLIP style loss (Figure (a)). Specifically, SGERA induces a noticeably smoother and better-conditioned optimization landscape compared to

Table 4. Distribution shift results (AUC). We bold the best results, and underline represents the second highest.

Source Domain Target Domain	Zero-shot	Training Data Ratio	PTBXL-Super CPSC2018	CPSC2018 CSN	PTBXL-Super CSN	CPSC2018 CSN	PTBXL-Super CSN	CPSC2018 CSN	Average
SimCLR (Chen et al., 2020)	✗	100%	69.62	73.05	56.65	66.36	59.67	62.11	65.22
BYOL (Grill et al., 2020)	✗		70.27	74.01	57.32	67.54	60.39	63.24	65.63
BarlowTwins (Zbontar et al., 2021)	✗		68.98	72.85	55.97	65.89	58.76	61.35	64.13
MoCo-v3 (Chen & He, 2021)	✗		69.41	73.29	56.54	66.12	59.82	62.07	64.21
SimSiam (Chen & He, 2021)	✗		70.06	73.92	57.21	67.48	60.23	63.09	65.33
TS-TCC (Eldele et al., 2021)	✗		71.32	75.16	58.47	68.34	61.55	64.48	66.55
CLOCS (Kiyasseh et al., 2021)	✗		68.79	72.64	55.86	65.73	58.69	61.27	63.83
ASTCL (Wang et al., 2023)	✗		69.23	73.18	56.61	66.27	59.74	62.12	64.19
CRT (Zhang et al., 2023)	✗		70.15	74.08	57.39	67.62	60.48	63.33	65.51
STMEM (Na et al., 2024)	✗		76.12	84.50	62.27	75.19	73.05	64.66	72.63
MERL (Liu et al., 2024)	✓	0%	<u>88.21</u>	78.01	76.77	76.56	74.15	<u>82.86</u>	79.42
DERI (Chen et al., 2025)	✓	0%	88.78	78.83	<u>79.50</u>	81.02	<u>76.70</u>	85.84	81.78
MELP (Wang et al., 2025a)	✓	0%	87.75	74.11	77.89	80.32	74.67	82.72	79.58
D-BETA (?)	✓	0%	72.09	79.11	77.12	82.91	76.24	80.10	77.92
SGERA	✓	0%	83.08	<u>79.77</u>	90.65	86.26	81.63	78.63	83.34

Table 5. Ablation study on different loss components.

$\mathcal{L}_{CLIP}$	$\mathcal{L}_{rbf}$	$\mathcal{L}_{KL}$	$\mathcal{L}_{dis}$	Zero-shot	Linear Probing (1%)
✓			✓	77.5	76.10
✓		✓		77.2	76.42
✓		✓	✓	78.3	77.37
	✓		✓	78.9	76.25
	✓	✓	✓	77.9	76.72
	✓	✓	✓	79.5	79.02

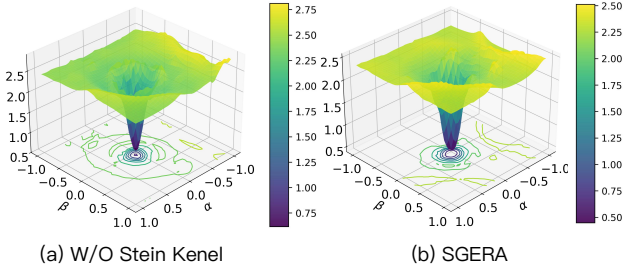


Figure 2. Loss landscape analysis.

its InfoNCE-based counterpart. The transition from a sharp, highly sensitive minimum to a flatter and more isotropic basin suggests improved robustness to parameter perturbations and more stable optimization dynamics. These geometric properties align with theoretical insights from kernel smoothing and Stein-based distribution alignment, helping explain the superior training stability and downstream generalization performance observed in practice.

4.4. Analysis of SGERA Framework

In this section, we conduct comprehensive ablation studies on the key components of the SGERA framework and then report the average performance of zero-shot classification and linear probing with 1% data across all datasets used. More details can be found in Appendix E.

4.4.1. ABLATION OF STEIN-GUIDED ALIGNMENT

To better verify the effect of our proposed Stein-Guided Alignment, we train the model with a different loss function as Table 5. This allows us to assess the contribution of each proposed loss. Replacing the CLIP-style contrastive loss with the proposed Stein-based RBF loss ($\mathcal{L}_{rbf}$) consistently improves zero-shot performance, highlighting the advantage of distribution-aware alignment over instance-level contrastive objectives. The distribution-level alignment term $\mathcal{L}_{dis}$ further enhances cross-modal robustness, particularly under the zero-shot setting, by explicitly matching global feature distributions. In addition, the KL regularization ($\mathcal{L}_{KL}$) stabilizes representation learning and improves linear probing performance. Overall, jointly optimizing $\mathcal{L}_{rbf}$, $\mathcal{L}_{dis}$, and $\mathcal{L}_{KL}$ achieves the best results, demonstrating the complementary benefits of instance-aware similarity modeling, distribution-level alignment, and regularization. Unlike InfoNCE-based CLIP loss, which performs instance-level alignment on a hyperspherical embedding space and is sensitive to sampling bias, the Stein-based RBF loss models cross-modal similarity through kernelized score matching, enabling alignment at the distribution level. This reduces representation bias induced by hard negatives and improves generality under domain shift by matching higher-order statistics rather than individual samples. As a result, the learned representations exhibit stronger robustness and transferability in zero-shot and low-resource settings.

4.4.2. ANALYSIS OF HARD SAMPLE MINING

To better evaluate the proposed hard sample mining, we design two variants: (1) **W/O ERM**, which removes the ERM task; (2) **Random Sampling**, which uses random sampling methods. Comparison results are shown in Figure 3. Hard Sampling (SGERA) consistently outperforms W/O ERM and Random Sampling across Zero-shot and Linear

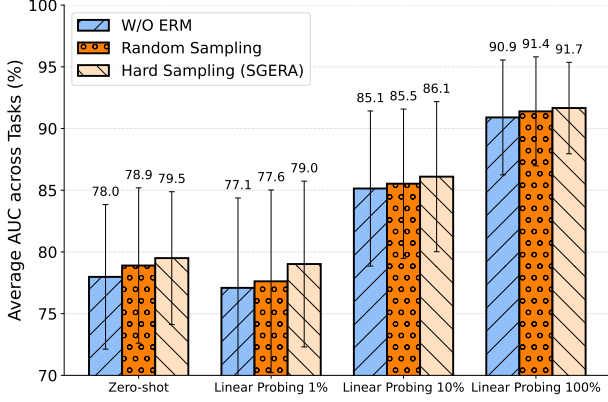


Figure 3. Analysis of Hard Sample Mining on downstream tasks.

Probing settings. In Zero-shot, it achieves 79.5% average accuracy, surpassing Random Sampling (79.1%) and W/O ERM (78.0%), demonstrating that selecting hard samples improves generalization. Under Linear Probing, performance increases with more labeled data, reaching 91.7% at 100%, with SGERA maintaining a consistent edge. Standard deviations decrease as the label ratio grows, indicating improved stability, and SGERA exhibits slightly lower variance than Random Sampling. These results suggest that SGERA not only enhances accuracy in low-resource scenarios but also maintains robustness with full supervision.

5. Conclusion

We analyze the limitations of CLIP-style loss in ECG-Report alignment and then propose SGERA, a novel framework leveraging Stein-kernel-based alignment for ECG-Report correspondence. By addressing modality heterogeneity through dual-level Stein kernel alignment and incorporating a hard sample mining strategy for ECG-Report matching, SGERA enhances instance-level discrimination. Extensive experiments on three public ECG benchmark datasets demonstrate that SGERA consistently outperforms existing baselines, validating its effectiveness in aligning ECG-Report pairs.

Impact Statement

SGERA focus on ECG-Report Alignment pretraining, which uses a large-scale public dataset: MIMIC-IV-ECG dataset. Although patient data is used, all the content related to patients’ personal information has been concealed, and we will not provide it. We believe that our research can further enhance the development of ECG signal diagnostic analysis, thereby promoting the progress of AI for Healthcare.

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References

- Chen, J., Hu, Y., Garg, L., Gadekallu, T. R., Srivastava, G., and Wang, W. Graph enhanced low-resource ecg representation learning for emotion recognition based on wearable internet of things. *IEEE Internet of Things Journal*, 2024.
- Chen, J., Dong, X., Wang, W., Zhou, S., Yu, L., and Hu, X. Deri: Cross-modal ecg representation learning with deep ecg-report interaction. In *34th International Joint Conference on Artificial Intelligence, IJCAI 2025*, pp. 4824–4832. International Joint Conferences on Artificial Intelligence, 2025.
- Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pp. 1597–1607. PmLR, 2020.
- Chen, X. and He, K. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 15750–15758, 2021.
- Cremer, C., Li, X., and Duvenaud, D. Inference suboptimality in variational autoencoders. In *International conference on machine learning*, pp. 1078–1086. PMLR, 2018.
- Eldele, E., Ragab, M., Chen, Z., Wu, M., Kwok, C. K., Li, X., and Guan, C. Time-series representation learning via temporal and contextual contrasting. *arXiv preprint arXiv:2106.14112*, 2021.
- Gow, B., Pollard, T., Nathanson, L. A., Johnson, A., Moody, B., Fernandes, C., Greenbaum, N., Waks, J. W., Eslami, P., Carbonati, T., et al. MIMIC-iv-ecg: Diagnostic electrocardiogram matched subset. *Type: dataset*, 6:13–14, 2023.
- Grill, J.-B., Strub, F., Althé, F., Tallec, C., Richemond, P., Buchatskaya, E., Doersch, C., Avila Pires, B., Guo, Z., Gheshlaghi Azar, M., et al. Bootstrap your own latent—a new approach to self-supervised learning. *Advances in neural information processing systems*, 33:21271–21284, 2020.
- Hu, R., Chen, J., and Zhou, L. Spatiotemporal self-supervised representation learning from multi-lead ecg

- signals. *Biomedical Signal Processing and Control*, 84: 104772, 2023.
- Huang, Y., Yen, G. G., and Tseng, V. S. Snippet policy network for multi-class varied-length ecg early classification. *IEEE Transactions on Knowledge and Data Engineering*, 35(6):6349–6361, 2022a.
- Huang, Y., Yen, G. G., and Tseng, V. S. Snippet policy network v2: Knee-guided neuroevolution for multi-lead ecg early classification. *IEEE Transactions on Neural Networks and Learning Systems*, 35(2):2167–2181, 2022b.
- Hung, M. P., Saeed, A., and Ma, D. Boosting masked ecg-text auto-encoders as discriminative learners. In *Forty-second International Conference on Machine Learning*, 2025.
- Jin, J., Wang, H., Li, H., Li, J., Pan, J., and Hong, S. Reading your heart: Learning ecg words and sentences via pre-training ecg language model. *arXiv preprint arXiv:2502.10707*, 2025.
- Jin, Q., Kim, W., Chen, Q., Comeau, D. C., Yeganova, L., Wilbur, W. J., and Lu, Z. Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval. *Bioinformatics*, 39(11):btad651, 2023.
- Kiyasseh, D., Zhu, T., and Clifton, D. A. Clocs: Contrastive learning of cardiac signals across space, time, and patients. In *International Conference on Machine Learning*, pp. 5606–5615. PMLR, 2021.
- Krishnan, R., Rajpurkar, P., and Topol, E. J. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering*, 6(12):1346–1352, 2022.
- Liu, C., Wan, Z., Ouyang, C., Shah, A., Bai, W., and Arducci, R. Zero-shot ecg classification with multimodal learning and test-time clinical knowledge enhancement. *arXiv preprint arXiv:2403.06659*, 2024.
- Liu, F., Liu, C., Zhao, L., Zhang, X., Wu, X., Xu, X., Liu, Y., Ma, C., Wei, S., He, Z., et al. An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection. *Journal of Medical Imaging and Health Informatics*, 8(7):1368–1373, 2018.
- Liu, Q. and Wang, D. Stein variational gradient descent: A general purpose bayesian inference algorithm. *Advances in neural information processing systems*, 29, 2016.
- Liu, W., Su, J., Chen, C., and Zheng, X. Leveraging distribution alignment via stein path for cross-domain cold-start recommendation. *Advances in Neural Information Processing Systems*, 34:19223–19234, 2021.
- Liu, W., Pan, S., Chang, S., Huang, Q., and Jiang, N. Self-supervised learning for electrocardiogram classification using lead correlation and decorrelation. *Applied Soft Computing*, 172:112871, 2025.
- McKeen, K., Oliva, L., Masood, S., Toma, A., Rubin, B., and Wang, B. Ecg-fm: An open electrocardiogram foundation model. *arXiv preprint arXiv:2408.05178*, 2024.
- Na, Y., Park, M., Tae, Y., and Joo, S. Guiding masked representation learning to capture spatio-temporal relationship of electrocardiogram. *arXiv preprint arXiv:2402.09450*, 2024.
- Oord, A. v. d., Li, Y., and Vinyals, O. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- Pham, M., Saeed, A., and Ma, D. C-melt: Contrastive enhanced masked auto-encoders for ecg-language pre-training. *arXiv e-prints*, pp. arXiv–2410, 2024.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PmLR, 2021.
- Siontis, K. C., Noseworthy, P. A., Attia, Z. I., and Friedman, P. A. Artificial intelligence-enhanced electrocardiography in cardiovascular disease management. *Nature Reviews Cardiology*, 18(7):465–478, 2021.
- Wagner, P., Strodthoff, N., Bousseljot, R.-D., Kreiseler, D., Lunze, F. I., Samek, W., and Schaeffter, T. Ptb-xl, a large publicly available electrocardiography dataset. *Scientific data*, 7(1):1–15, 2020.
- Wang, F., Xu, J., and Yu, L. From token to rhythm: A multi-scale approach for ecg-language pretraining. In *Forty-second International Conference on Machine Learning*, 2025a.
- Wang, M., Xiao, Y., Wang, B., Zhang, S., Ye, S., Wang, W., Yin, H., Guo, R., and Xu, Z. Findrec: Stein-guided entropic flow for multi-modal sequential recommendation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pp. 3008–3018, 2025b.
- Wang, N., Feng, P., Ge, Z., Zhou, Y., Zhou, B., and Wang, Z. Adversarial spatiotemporal contrastive learning for electrocardiogram signals. *IEEE Transactions on Neural Networks and Learning Systems*, 35(10):13845–13859, 2023.
- Wen, L., Zhou, Y., He, L., Zhou, M., and Xu, Z. Mutual information gradient estimation for representation learning. *arXiv preprint arXiv:2005.01123*, 2020.

- Wen, L., Wang, X., Liu, J., and Xu, Z. Mveb: Self-supervised learning with multi-view entropy bottleneck. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(9):6097–6108, 2024.
- Xiao, Y., Wang, S., Wang, B., Zhang, Z., Zhang, Y., Liu, S., Feng, C., Li, X., and Zhuang, F. Mars: Modality-aligned retrieval for sequence augmented ctr prediction. *arXiv preprint arXiv:2509.01184*, 2025.
- Yu, H., Guo, P., and Sano, A. Ecg semantic integrator (esi): A foundation ecg model pretrained with llm-enhanced cardiological text. *arXiv preprint arXiv:2405.19366*, 2024.
- Yuan, X., Wang, W., Liu, H., Chen, J., and Hu, X. Ecg2tok: Ecg pre-training with self-distillation semantic tokenizers. In *34th International Joint Conference on Artificial Intelligence, IJCAI 2025*, pp. 9990–9998. International Joint Conferences on Artificial Intelligence, 2025.
- Zbontar, J., Jing, L., Misra, I., LeCun, Y., and Deny, S. Barlow twins: Self-supervised learning via redundancy reduction. In *International conference on machine learning*, pp. 12310–12320. PMLR, 2021.
- Zhang, W., Yang, L., Geng, S., and Hong, S. Self-supervised time series representation learning via cross reconstruction transformer. *IEEE Transactions on Neural Networks and Learning Systems*, 2023.
- Zheng, J., Guo, H., and Chu, H. A large scale 12-lead electrocardiogram database for arrhythmia study (version 1.0. 0). *PhysioNet 2022* Available online http://physionet.org/content/ecg_arrhythmia10_0 accessed on, 23:7, 2022.

A. Theoretical Analysis on ECG-Report Alignment

In this section, we provide a theoretical analysis of ECG-report alignment, explaining why Stein-based alignment offers a more suitable and effective solution.

A.1. Limitations of CLIP-style InfoNCE under Modality Heterogeneity

CLIP-style contrastive learning aligns multimodal representations by maximizing the similarity between paired samples while contrasting against other samples in the batch. Formally, given paired ECG-report embeddings (z_i^E, z_i^R) from ECG and report encoders, the similarity matrix via inner product is computed as $S_{ij} = \langle z_i^E, z_j^R \rangle$, and optimized via a symmetric InfoNCE loss as Eq. 1.

This formulation implicitly assumes that each instance can be represented as a single point in a shared embedding space. Under this assumption, InfoNCE can be shown to provide a lower bound on the mutual information (MI) between two random variables (Oord et al., 2018), as Eq. 17:

$$I(X, Y) \geq \log(N) - \mathcal{L}_{\text{InfoNCE}}. \quad (17)$$

In addition, there are no explicit random variables, only one deterministic embedding, which means that given x , the embedding z is deterministic, as

However, this point-wise assumption becomes problematic in ECG-Report alignment due to severe modality heterogeneity:

- ECG signals are continuous, temporal, and morphology-driven, where a single recording may exhibit multiple clinically relevant patterns.
- Clinical reports are discrete, compositional, and semantic, where the same ECG may correspond to multiple valid textual descriptions.

As a result, a single embedding vector is insufficient to characterize the underlying semantics of an instance. Treating ECG-Report pairs as point variables leads to unstable alignment and exacerbates the false-negative issue, where semantically related pairs are incorrectly pushed apart. Under this formulation, the alignment objective should measure distributional consistency rather than point-wise similarity.

A.2. Log-RBF Stein Kernel as Distributional Similarity

To quantify the similarity between two distributions, we adopt a log-RBF kernel defined over embedding pairs as Eq 3. The bandwidth σ in Eq 3 is adaptively estimated from the data using the median heuristic during training, ensuring robustness across heterogeneous embedding scales. Importantly, the kernel output is already in log-space, avoiding the numerical instability and over-sharpening that may arise from directly applying an exponential RBF kernel.

Unlike dot-product similarity, the RBF kernel corresponds to an inner product in Reproducing kernel Hilbert space (RKHS), as Eq 18:

$$k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathcal{H}}, \quad (18)$$

Thus, the kernel matrix K^{rbf} captures higher-order moment matching, not just mean alignment. We replace the cosine similarity matrix in CLIP with the log-RBF kernel matrix K^{rbf} as Eq. 4. This formulation preserves the computational structure and optimization benefits of InfoNCE, while replacing point-wise similarity with a kernel-induced energy function. The log-RBF kernel naturally arises in kernelized Stein discrepancies, which measure divergence between two distributions without requiring explicit density estimation. In this way, it transforms InfoNCE into minimizing a Stein discrepancy-induced contrastive objective, aligning distributions rather than points.

A.3. Comparison Summary

In a summary, CLIP-style InfoNCE maximizes a lower bound of mutual information between point-mass variables, i.e., $I(Z^E, Z^R)$, where Z^E and Z^R are treated as Dirac random variables. Stein-based InfoNCE does not correspond to a mutual information bound; instead, it minimizes a kernelized discrepancy between conditional embedding distributions, aligning $p(z^E|x^E)$ and $p(z^R|x^R)$ in a Reproducing Kernel Hilbert Space (RKHS). Figure 4 provides a better understanding of our

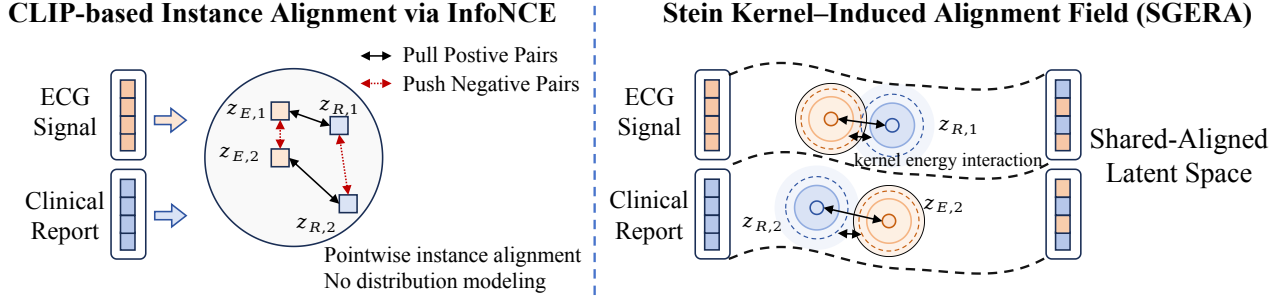


Figure 4. CLIP vs. Stein-guided ECG-report alignment. CLIP aligns ECG and report representations by maximizing cosine similarity between paired instances in a shared hyperspherical latent space using the InfoNCE loss. This formulation operates at the pointwise instance level and treats all other samples as negatives, without explicitly modeling cross-modal distributional structure. In contrast, SGERA replaces cosine similarity with a Stein-guided RBF kernel, inducing an alignment field in a shared-aligned latent space. Representations interact through local kernel influence regions, and modality-specific kernel structures jointly guide instance- and distribution-level alignment.

proposed SGERA method. As illustrated in Figure 4, SGERA transforms the alignment task from a zero-sum contrastive game between points into a harmonized flow between distributions. While CLIP enforces pointwise instance alignment through rigid cosine similarity, it neglects the underlying manifold geometry, leading to instability under cross-modal distribution shifts. In contrast, SGERA introduces a Stein-induced alignment field that operates hierarchically:

- Instance-level consistency is achieved through local kernel energy interactions (represented by the overlapping influence regions), which provide robust, non-linear pairwise supervision;
- Distribution-level synchronization is guided by modality-specific kernel structures (indicated by the dashed manifold contours), which capture higher-order statistics to align the global latent geometry of ECG and text. This dual-level synergy ensures that SGERA learns physiologically meaningful representations that are both discriminative at the sample level and structurally coherent across modalities.

When the paired modalities exhibit intrinsic ambiguity or multi-modality—such as ECG signals and clinical reports—Stein-based InfoNCE provides a more appropriate alignment objective than CLIP-style InfoNCE, as it:(1) avoids Dirac collapse, (2) reduces false-negative repulsion, and (3) captures higher-order semantic consistency across modalities. Therefore, CLIP-style InfoNCE maximizes mutual information between deterministic point embeddings, whereas Stein-based InfoNCE minimizes a kernelized discrepancy between conditional embedding distributions, making it better suited for heterogeneous and ambiguous multimodal alignment.

A.4. Complete Derivation of Gradient from Stein-Kernel

To better illustrate the RBF-Stein kernel for instance-level alignment, we provided a complete derivation of Eq. 6.

1. Following Eq. 3 in the manuscript, the instance-level alignment loss (considering the first symmetric term) is defined via the RBF-Stein kernel matrix K :

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(K_{ii}/\tau)}{\sum_{j=1}^N \exp(K_{ij}/\tau)} \quad (19)$$

where the kernel elements K_{ij} are computed based on the negative squared Euclidean distance:

$$K_{ij} = -\frac{\|z_E^i - z_R^j\|^2}{2\sigma^2} \quad (20)$$

2. Expanding the logarithm for a single anchor i yields the individual loss component ℓ_i :

$$\ell_i = -\frac{1}{N} \left[\frac{K_{ii}}{\tau} - \log \left(\sum_{j=1}^N \exp \left(\frac{K_{ij}}{\tau} \right) \right) \right] \quad (21)$$

3. Let $\tilde{p}_{ij}$ denote the softmax-normalized similarity for the i -th row, and δ_{ij} be the Kronecker delta:

$$\tilde{p}_{ij} = \frac{\exp(K_{ij}/\tau)}{\sum_{k=1}^N \exp(K_{ik}/\tau)} \quad (22)$$

4. The derivative of $\mathcal{L}$ with respect to the kernel element K_{ij} simplifies using $\tilde{p}_{ij}$:

$$\frac{\partial \mathcal{L}}{\partial K_{ij}} = \frac{\partial \ell_i}{\partial K_{ij}} = -\frac{1}{N} \left[\frac{1}{\tau} \delta_{ij} - \frac{\exp(K_{ij}/\tau)}{\sum_{k=1}^N \exp(K_{ik}/\tau)} \cdot \frac{1}{\tau} \right] = \frac{1}{N\tau} (\tilde{p}_{ij} - \delta_{ij}) \quad (23)$$

5. From the definition of K_{ij} , the gradient with respect to the embedding z_E^i is:

$$\nabla_{z_E^i} K_{ij} = \nabla_{z_E^i} \left(-\frac{1}{2\sigma^2} (z_E^i - z_R^j)^T (z_E^i - z_R^j) \right) = \frac{z_R^j - z_E^i}{\sigma^2} \quad (24)$$

6. Applying the chain rule $\nabla_{z_E^i} \mathcal{L} = \sum_j \frac{\partial \mathcal{L}}{\partial K_{ij}} \nabla_{z_E^i} K_{ij}$:

$$\nabla_{z_E^i} \mathcal{L} = \sum_{j=1}^N \left[\frac{1}{N\tau} (\tilde{p}_{ij} - \delta_{ij}) \right] \left[\frac{z_R^j - z_E^i}{\sigma^2} \right] \quad (25)$$

7. By extracting the positive constant $\frac{1}{N\tau\sigma^2}$, the final gradient expression is:

$$\nabla_{z_E^i} \mathcal{L} \propto \sum_{j=1}^N (z_R^j - z_E^i) (\tilde{p}_{ij} - \delta_{ij}) \quad (26)$$

B. Details about ECG-Report Matching Task

Sampling Details. For hard negative mining, we adopt a similarity-proportional stochastic sampling strategy instead of deterministically selecting the single most similar negative. Specifically, given a batch of ECG embeddings $\{z_{i,E}\}_{i=1}^B$, we compute the intra-modal similarity matrix as

$$s_{ij}^e = \frac{z_{i,E}^\top z_{j,E}}{\tau_e}, \quad i \neq j,$$

where τ is the temperature parameter, which is set as 0.07. To avoid trivial self-matching, diagonal entries are masked out. The similarity scores are then normalized with a softmax function to obtain a categorical distribution

$$p_{ij}^e = \frac{\exp(s_{ij}^e)}{\sum_{k \neq i} \exp(s_{ik}^e)}.$$

A hard ECG negative is sampled from this distribution using multinomial sampling, which assigns higher probabilities to semantically similar yet mismatched samples while retaining stochasticity.

The same procedure is applied symmetrically to text embeddings using temperature τ . To further capture cross-modal confusions, we compute cross-modal similarity matrices between ECG and text embeddings and sample hard negatives across modalities using the same similarity-proportional strategy with temperature τ . In total, for each positive ECG-report pair, we obtain intra-modal and cross-modal hard negatives from both modalities, forming a diverse and informative negative set for the auxiliary ECG-Report Matching task.

Similarity-Proportional Sampling Motivation. Instead of selecting the single hardest negative, we sample negatives with probabilities proportional to their similarity scores. This design avoids overfitting to extreme outliers and mitigates false-negative risks, especially under noisy ECG morphologies and ambiguous clinical descriptions. In addition, in ECG-report alignment, samples with high similarity may still share underlying clinical semantics. Our soft hard-negative sampling strategy reduces the risk of treating such samples as strict negatives, which is crucial in medical scenarios with subtle morphological variations.

ERM Architecture Details. The fusion module FuseAttn consists of a single-layer self-attention with hidden dimension d , followed by mean pooling. The matching head $g(\cdot)$ is a two-layer MLP with ReLU activation.

Table 6. Distribution shift category matching relations.

Target Category	PTBXL-Super	CPSC2018	CSN
AFIB	-	AFIB	AFIB
APB	-	PAC	-
CLBBB	CD	-	-
CRBBB	CD	-	-
HYP	HYP	-	RVH, LVH
LBBS	CD	-	CLBBB
MI	MI	-	MI
NORM	NORM	NORM	SR
PAC	-	PAC	-
SR	-	NORM	SR
STD	STTC	STD	STE, STTC, STTU, STDD
STE	STTC	STE	STE
STTC	STTC	-	STTC, STE, TWO, STTU, QTIE, TWC
VPC / VPB	-	VPC	VPB
1AVB	CD	1AVB	1AVB
2AVB / AVB	-	-	2AVB, 2AVB1, AVB
RBBB	CD	CRBBB	RBBB

C. Dataset Details

For pretraining, we use the MIMIC-IV-ECG v1.0 dataset (Gow et al., 2023). Specifically, we adopt the same preprocessing as MERL (Liu et al., 2024). Specifically, all the samples without reports containing more than 3 words are removed. All the "NaN" or "inf" in the ECG signal recordings are replaced with mean interpolation.

When conducting downstream tasks, we use three datasets: PTB-XL (Wagner et al., 2020), CPSC2018 (Liu et al., 2018), and CSN (Zheng et al., 2022). For the PTB-XL dataset, depending on the granularity of the annotation, it contains four subgroups for multi-label classification: Superclass with 5 categories, Subclass with 23 categories, Form with 12 categories, and Rhythm with 12 categories. We split the dataset in the same way as the official strategy (Wagner et al., 2020).

For the CPSC2018 dataset, we randomly split the data into train, validation, and test in a 7: 1:2 ratio.

For the CSN dataset, the original dataset contains 45,152 samples. Following previous works, we remove the signals with "unknown" annotation. Finally, 23,026 ECG records with 38 categories are split as 7:1:2.

D. Implementation Details on Downstream Tasks

After pretraining, we use three downstream evaluation tasks to verify the model’s performance: zero-shot classification, linear probing, and distribution shift evaluation.

For **zero-shot classification**, we adopt the same pipeline and category prompt as MERL (Liu et al., 2024). This is a retrieval-based classification task, which means that we conduct classification based on the similarity between the latent embeddings of ECG signals and all the category prompt embeddings. Specifically, given ECG signals, we first use the pre-trained model to obtain the embeddings z_E . For category prompt embeddings, the text encoder used for alignment in pretraining is adopted for z_C . We then calculate the similarity between z_E and z_C and retrieve the one with the greatest similarity as the predicted category.

For **Linear Probing**, we transfer the pretrained ECG model and freeze all the parameters. We then add a trainable linear classification head on the top of the model. We then adopt the AdamW optimizer with an initial learning rate of $5e-4$ and a weight decay of $5e-4$. The model is trained for 100 epochs with a batch size of 16.

For **Distribution Shift**, we adopt the same way as other zero-shot methods, which compute the metric across the overlapped categories, and the category matching relations are shown in Table 6. Specifically, for each target category, we computed the average AUC across all corresponding source categories as a metric, consistent with other works.

The baselines we compared include single-model supervised learning methods: SimCLR (Chen et al., 2020), BYOL (Grill

et al., 2020), BarlowTwins (Zbontar et al., 2021), MoCo-v3 (Chen & He, 2021), SimSiam (Chen & He, 2021), TS-TCC (Eldele et al., 2021), CLOCS (Kiyasseh et al., 2021), ASTCL (Wang et al., 2023), CRT (Zhang et al., 2023), STMEM (Na et al., 2024), Heartlang (Jin et al., 2025). We also compare our proposed SGERA with multi-modal ECG representation learning methods, including: MERL (Liu et al., 2024), DERI (Chen et al., 2025), MELP (Wang et al., 2025a), D-BETA (?).

E. More Details about Ablation Study Results

E.1. Analysis of SGERA Framework

To better illustrate the effect of the proposed SGERA framework, we provide more details in this section. Table 7 shows the details of zero-shot classification to analyze the proposed Stein-Guided Alignment loss. We can see that, when the model adopt the $\mathcal{L}_{rbf}$ to replace $\mathcal{L}_{CLIP}$ and keep others the same, the average performance shows a great improvement (77.5 to 78.9, 77.2 to 77.9, and 78.3 to 79.5). And we can also see that after removing the $\mathcal{L}_{dist}$, the model’s performance will have an obvious decline.

Table 7. Ablation study of different loss components on zero-shot classification performance.

$\mathcal{L}_{CLIP}$	$\mathcal{L}_{rbf}$	$\mathcal{L}_{KL}$	$\mathcal{L}_{dist}$	PTBXL-Super	PTBXL-Sub	PTBXL-Form	PTBXL-Rhythm	CPSC2018	CSN	Average
✓			✓	77.6	80.6	69.3	78.1	82.4	76.8	77.5
✓		✓		78.5	79.1	67.3	80.5	82.2	75.5	77.2
✓		✓	✓	80.2	81.3	66.2	82.0	84.3	76.0	78.3
	✓		✓	78.4	78.1	67.6	85.8	86.0	77.3	78.9
	✓	✓		77.4	79.5	67.3	84.4	83.0	75.5	77.9
	✓	✓	✓	78.8	79.5	71.3	83.3	87.1	77.3	79.5

Table 8. Ablation study of different loss components with varying labeled data ratios.

Method	PTBXL-Super			PTBXL-Sub			PTBXL-Form			PTBXL-Rhythm			CPSC2018			CSN		
	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%
$\mathcal{L}_{CLIP} + \mathcal{L}_{KL}$	85.26	88.89	91.89	77.12	84.02	88.25	64.15	73.46	84.78	77.44	88.72	94.23	79.24	89.67	95.12	75.32	86.37	94.12
$\mathcal{L}_{CLIP} + \mathcal{L}_{dist}$	85.34	89.80	91.79	76.05	84.08	88.46	63.09	73.76	85.76	78.46	89.79	94.81	79.79	88.51	95.36	73.88	86.22	94.65
$\mathcal{L}_{CLIP} + \mathcal{L}_{KL} + \mathcal{L}_{dist}$	86.27	89.66	91.51	78.64	83.97	88.19	65.40	73.04	84.38	77.38	88.04	94.07	79.18	90.53	95.46	77.37	86.71	93.98
$\mathcal{L}_{rbf} + \mathcal{L}_{dist}$	85.66	89.73	91.80	78.57	83.69	88.22	64.46	75.81	85.12	75.60	86.77	95.28	79.47	89.20	95.35	73.73	83.85	94.94
$\mathcal{L}_{rbf} + \mathcal{L}_{KL}$	85.95	90.09	91.73	79.87	81.78	88.53	62.75	77.70	85.22	76.02	90.88	95.07	79.50	91.00	95.61	76.25	86.76	94.68
SGERA	86.44	89.81	91.76	80.50	83.94	89.25	66.21	75.30	85.55	80.68	91.05	93.68	80.91	91.13	95.58	79.37	86.44	94.12

We then also provided the result details under linear probing setting as Table 8. We can see that the complete SGERA method achieve the best performance among almost all dataset when training with 1% data. This observation indicates that SGERA learns more informative and transferable representations during pre-training, which can be effectively leveraged with minimal supervision. Such behavior is particularly desirable in realistic clinical settings, where annotated ECG data are often scarce or expensive to obtain.

Table 9. Zero-shot classification performance (AUC) across datasets. The **best** and second best results are highlighted.

Method	PTBXL-Super	PTBXL-Sub	PTBXL-Form	PTBXL-Rhythm	CPSC2018	CSN	Average
w/o ERM	77.6	77.9	68.1	83.9	84.0	76.4	78.0
Random Sample	<u>78.6</u>	80.1	<u>68.3</u>	<u>83.7</u>	<u>85.9</u>	<u>76.9</u>	<u>78.9</u>
Hardest Sample	78.1	79.5	66.5	83.7	80.3	77.8	77.7
SGERA	78.8	<u>79.5</u>	71.3	83.3	87.1	77.3	79.5

In addition, to empirically validate the intrinsic effectiveness of each proposed loss component, we adopt a parameter-free integration strategy by directly aggregating all loss terms with unit weights. Although further performance gains might be achievable through meticulous weight tuning, we intentionally bypass such heuristics to eliminate the influence of additional hyperparameters. This ensures that the observed improvements are strictly attributable to our methodological design rather than the artifacts of exhaustive hyperparameter optimization.

Table 10. Linear probing results (AUC) with different sampling strategies under varying labeled data ratios.

Method	PTBXL-Super			PTBXL-Sub			PTBXL-Form			PTBXL-Rhythm			CPSC2018			CSN		
	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%	1%	10%	100%
w/o L_{ERM}	85.57	89.65	91.85	80.38	82.28	88.18	63.63	73.84	82.81	77.25	89.65	93.03	78.02	89.67	95.00	77.69	85.77	94.54
Random Sample	87.19	89.96	91.72	80.44	84.40	87.98	64.62	74.47	84.28	79.36	90.28	94.47	77.95	89.53	95.40	76.16	84.51	94.50
Hardest Sample	84.93	88.29	90.23	79.08	82.13	86.97	63.72	73.28	82.79	76.71	88.80	92.14	77.62	88.59	93.72	76.42	84.13	92.79
SGERA	86.44	89.81	91.76	80.50	83.94	89.25	66.21	75.30	85.55	80.68	91.05	93.68	80.91	91.13	95.58	79.37	86.44	94.12

E.2. Analysis of Hard Sample Mining

To better illustrate the effect of the proposed hard sample mining for ECG-Report Matching task as auxiliary task for better representation learning, we provide more details of results here. As shown in Table 9, the inclusion of hard sample mining through SGERA consistently improves zero-shot performance across most datasets. Compared with random sampling and the baseline without ERM, SGERA achieves higher AUC scores on PTBXL-Form, CPSC2018, CSN, and the overall average. While SGERA does not achieve the best result on every single dataset, its overall performance demonstrates that leveraging hard samples helps the model learn more discriminative ECG-report representations, which are beneficial for downstream classification tasks with limited supervision. We also find that when we adopt hardest sampling, the performance of the model is even lower than that of the model without ERM tasks. It illustrates that hardest-only mining leads to unstable training and inferior performance.

As shown in Table 10, SGERA demonstrates consistently strong performance across most datasets and data scales. Notably, it achieves the highest or near-highest AUC scores under the 1% linear probing setting, indicating that the learned representations are particularly effective in extremely low-resource scenarios. However, SGERA does not always achieve the best performance at larger data ratios (10% or 100%), reflecting that other sampling strategies can be competitive depending on the dataset and label availability. It should also be noted that when we adopt the hardest sampling strategy, the performance of the model has deteriorated, even falling below that achieved when the auxiliary task was removed. This further validates our observation that the proposed hardest sample mining approach leads to similar samples being strictly classified as negative examples, thereby causing representational collapse.

Overall, these results suggest that SGERA enhances representation quality and label efficiency while maintaining robust performance across different downstream tasks.

F. Extensibility Analysis

To better discuss the extensibility of the proposed dual-level Stein-Guided Alignment method, we conduct more experiments. Specifically, we apply the proposed alignment loss to the existing baselines, including MERL (Liu et al., 2024) and DERI (Chen et al., 2025). For each baseline, we keep the model components unchanged while changing the optimization objective to get two variants. MERL-Ins and DERI-Ins mean that we use the proposed RBF-Stein Kernel for Instance-Level Alignment to replace the original alignment loss. Then, MERL-Stein and DERI-Stein mean to use the proposed dual-level Stein-guided alignment loss (both $\mathcal{L}_{inst}$ and $\mathcal{L}_{dist}$) to replace the original alignment loss. We also counted the number of training parameters of the model at the same time to assess whether our loss would lead to greater computational costs. Experimental results are shown in Figure 5.

As illustrated in Figure 5, SGERA consistently delivers competitive results in both zero-shot (79.5% AUC) and linear probing (91.66% AUC) tasks, underscoring its capacity to extract robust, domain-agnostic cross-modal representations. Remarkably, SGERA achieves linear probing performance that rivals or even surpasses much larger models,

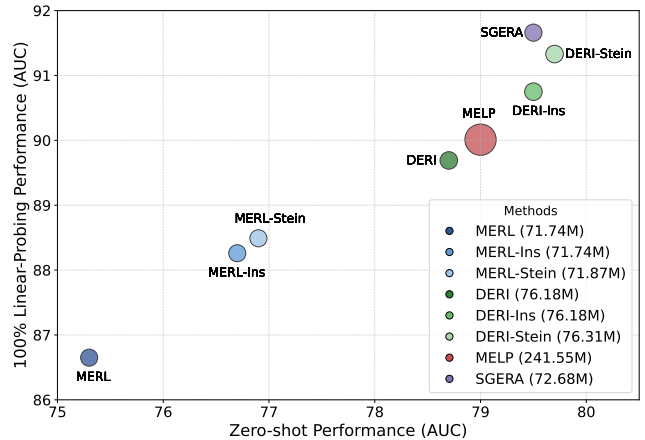


Figure 5. Zero-shot vs Linear-Probing Performance Comparison. We apply the proposed Stein-Guided Alignment loss on existing baselines and compare the performance and trainable parameter number.

such as MELP (241.55M parameters). This parity demonstrates that SGERA is highly parameter-efficient, achieving state-of-the-art diagnostic accuracy without the overhead of massive architectural scaling.

We also find that methods incorporating intra-modal or Stein-based alignments (MERL-Stein, DERI-Stein, etc.) improve both Zero-shot and Linear-Probing performance over their base variants. This indicates that our proposed dual-level Stein-Guided Alignment strategies not only enhance generalization to unseen data but also facilitate more effective feature extraction when labeled data is available. In particular, the Linear-Probing results show clear gains, suggesting that these strategies strengthen the learned embeddings in a way that benefits downstream fine-tuning.

Finally, while SGERA generally performs best overall, it is not strictly superior in every setting. DERI-Stein achieves slightly higher Zero-shot AUC in certain cases with almost 4M additional parameters. Overall, the results support that SGERA effectively improves both zero-shot generalization and fine-tuning performance while remaining parameter-efficient, with gains most evident under limited data or challenging alignment conditions. It also shows that the designed dual-level Stein-guided alignment loss (both $\mathcal{L}_{inst}$ and $\mathcal{L}_{dist}$) can be effectively adopted to other methods for performance improvement, which shows great extensibility.

G. Multi-modal Representation Analysis

To better discuss the effect of SGERA for multi-modal representations alignment, we conduct several analysis here. We adopt the completed SGERA and its two variants: W/O DIS which removes the $\mathcal{L}_{dist}$ and W/O RBF which uses the $\mathcal{L}_{CLIP}$ to replace $\mathcal{L}_{rbf}$. We conduct t-SNE visualization on the learned ECG and report representations first and then draw circles based on the average intra-cluster distance of each cluster and the cluster center to represent the distribution area characterized by this modality. Visualization results are shown in Figure 6.

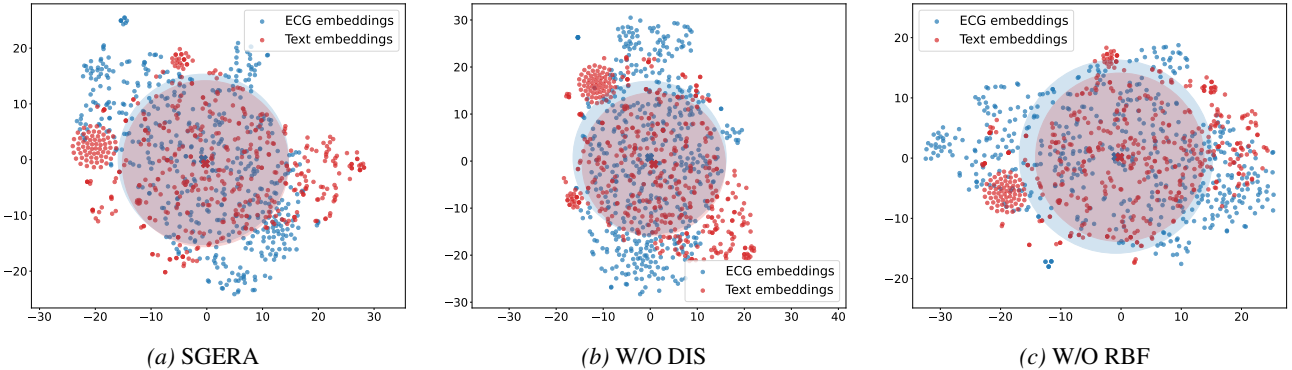


Figure 6. T-SNE visualization of multi-modal representations from different methods on MIMIC-IV ECG dataset.

As shown in Figure 6 presents the t-SNE visualizations of joint ECG-report embeddings, providing a qualitative assessment of alignment quality. As observed in Figure 6a, the representations learned by SGERA exhibit a highly integrated and dense distribution, where the ECG and text modalities show extensive overlap with blurred boundaries. This pattern indicates that SGERA successfully synchronizes the two heterogeneous manifolds into a unified latent space.

In contrast, the ablated versions, W/O DIS (Figure 6b) and W/O RBF (Figure 6c), show noticeably inferior alignment. Specifically, removing the distribution-level alignment (W/O DIS) results in looser embedding clusters and significant modality separation, highlighting the critical role of the Stein-Score kernel in bridging global structural gaps. Similarly, the absence of the Stein-RBF kernel (W/O RBF) leads to more discrete and fragmented points, suggesting a loss of fine-grained pairwise consistency. These visual comparisons reinforce that the synergy between instance-level and distribution-level alignment is essential for learning robust, cross-modal cardiac representations.

To quantify cross-modal distribution alignment in the 2D visualization space, we compute the overlap ratio between the ECG and report embedding clusters. Specifically, for each modality, we estimate the cluster center as the mean of the corresponding t-SNE embeddings, and define the cluster radius as the average Euclidean distance from samples to the center. Given two circles with centers $\mathbf{c}_1, \mathbf{c}_2$ and radii r_1, r_2 , the overlap area is computed analytically based on their geometric

intersection. The overlap ratio is then defined as:

$$\text{Overlap Ratio} = \frac{\mathcal{A}_{\text{overlap}}}{\min(\mathcal{A}_1, \mathcal{A}_2)}, \quad (27)$$

where $\mathcal{A}_{\text{overlap}}$ denotes the intersection area of the two circles. This metric reflects the extent to which the two modality distributions occupy a shared region, complementing instance-level similarity measures. The results are shown in Figure 7.

While SGERA yields the highest average cosine similarity and overlap ratio among the evaluated variants, these metrics warrant a nuanced interpretation. Such improvements do not signify an absolute convergence of ECG signals and clinical reports; rather, they reflect a heightened propensity toward both instance-level and distribution-level coherence. Furthermore, as the overlap ratio is derived from low-dimensional t-SNE projections, it primarily captures topological structural alignment within the visualization manifold, which may not fully encapsulate the intricate, high-dimensional geometric relationships inherent in the original embedding space.

Nevertheless, the congruence between high-dimensional cosine similarity and overlap metrics suggests that the observed performance gains are systematic rather than artifacts of a specific evaluation protocol. Compared to ablated variants, SGERA achieves a more stable cross-modal alignment by effectively mitigating distributional mismatch, even though certain inherent ambiguities between modalities persist. These findings indicate that while SGERA significantly enhances alignment quality, it does so in a principled and incremental manner, acknowledging that some degree of cross-modal discrepancy remains inherent to the task.

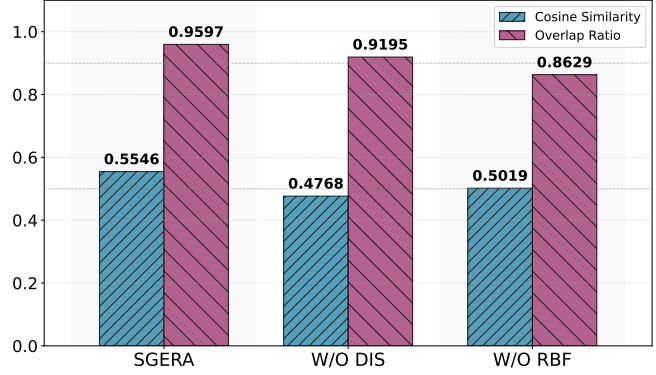


Figure 7. Average cosine similarity of all ECG-Report pairs and overlap ratio of modality distribution circle.

H. Limitation and Future Directions.

SGERA is designed to primarily address modality heterogeneity between ECG signals and clinical reports, focusing on robust cross-modal alignment at the semantic level. As a consequence, the current framework does not explicitly target fine-grained feature modeling within each modality. We consider this a complementary research direction rather than a limitation of the proposed design. In future work, integrating fine-grained ECG morphological cues and more detailed clinical language representations could further enrich the learned embeddings, potentially benefiting interpretability-oriented or diagnosis-specific downstream applications.

I. The Usage of large language models LLM

We employed large language models (LLMs) in a limited capacity, strictly for writing assistance tasks such as grammar correction, style refinement, and table formatting. All main model implementations, data processing pipelines, and training routines were entirely written and verified by the authors. All suggested edits were carefully reviewed and selectively incorporated by the authors. The scientific content, ideas, analyses, and conclusions presented in this paper are entirely our own. The authors take full responsibility for the paper’s content, including any remaining errors or inaccuracies.